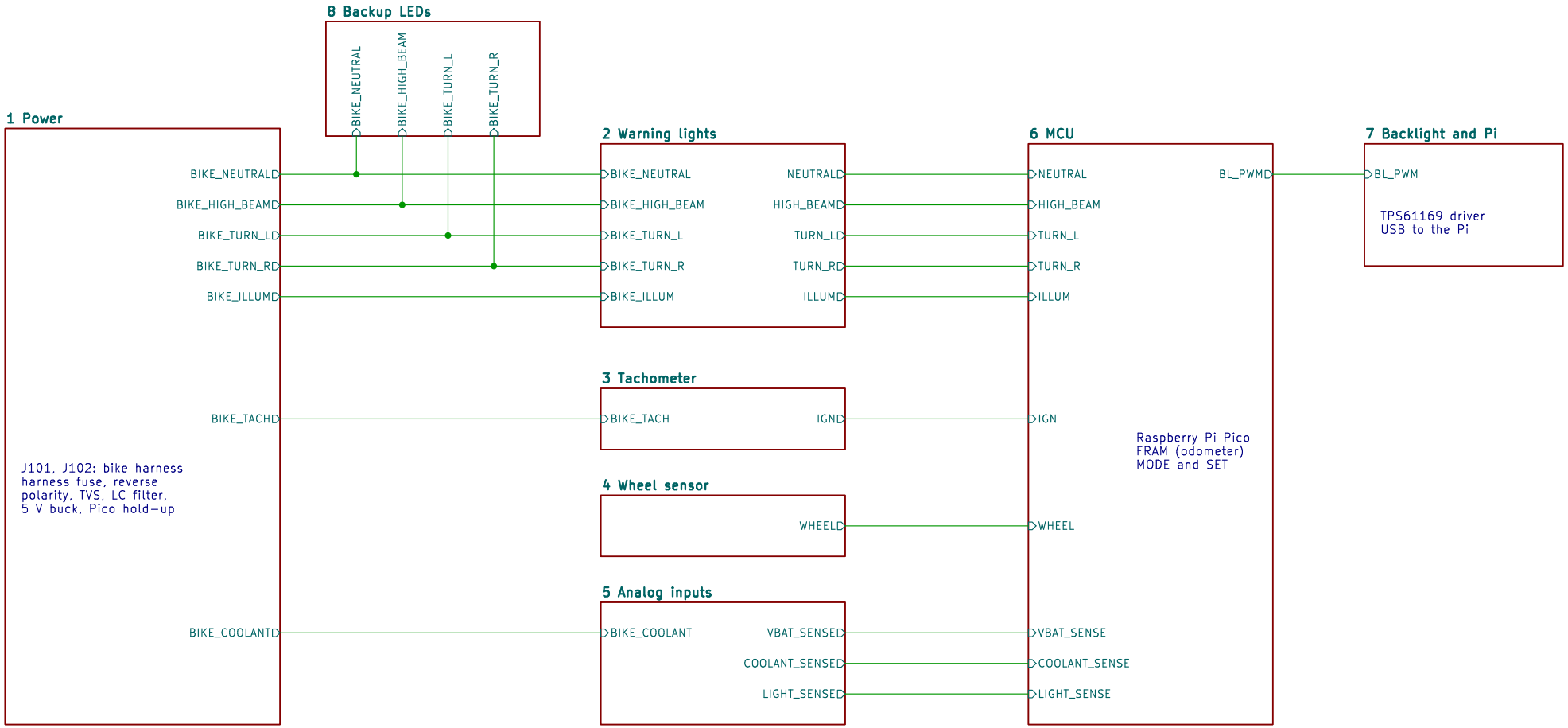
tdm-dash sensor module: block diagram



Each block is one sheet; the wires between the blocks are the signals that cross sheets.

Sheets

1 Power: bike connectors J101 and J102, harness fuse, reverse polarity, TVS, LC filter, 5 V buck, Pico hold-up
2 Warning lights: neutral, high beam, turn signals and illumination through EL817 optocouplers
3 Tachometer: coil negative to the H11L1 Schmitt trigger
4 Wheel sensor: LJ12A3 inductive sensor on J401, through an EL817
5 Analog inputs: battery voltage, coolant sender and ambient light to the ADC
6 MCU: Raspberry Pi Pico, FRAM for the odometer, MODE and SET buttons
7 Backlight and Pi: TPS61169 backlight driver dimmed by the Pico, USB link to the Pi
8 Backup LEDs: indicators wired straight to the bike, alive with the electronics off

Net names

BIKE_... wire from the bike harness: up to 14.5 V, with spikes
+12V_P protected 12 V · +5V buck output · VSYS_PICO Pico supply · +3V3 Pico regulator
Supplies are global power symbols and are not drawn on this page.

Orange notes marked [TO VERIFY] wait for the bike measurements (docs/research/original-wiring.md).

github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand

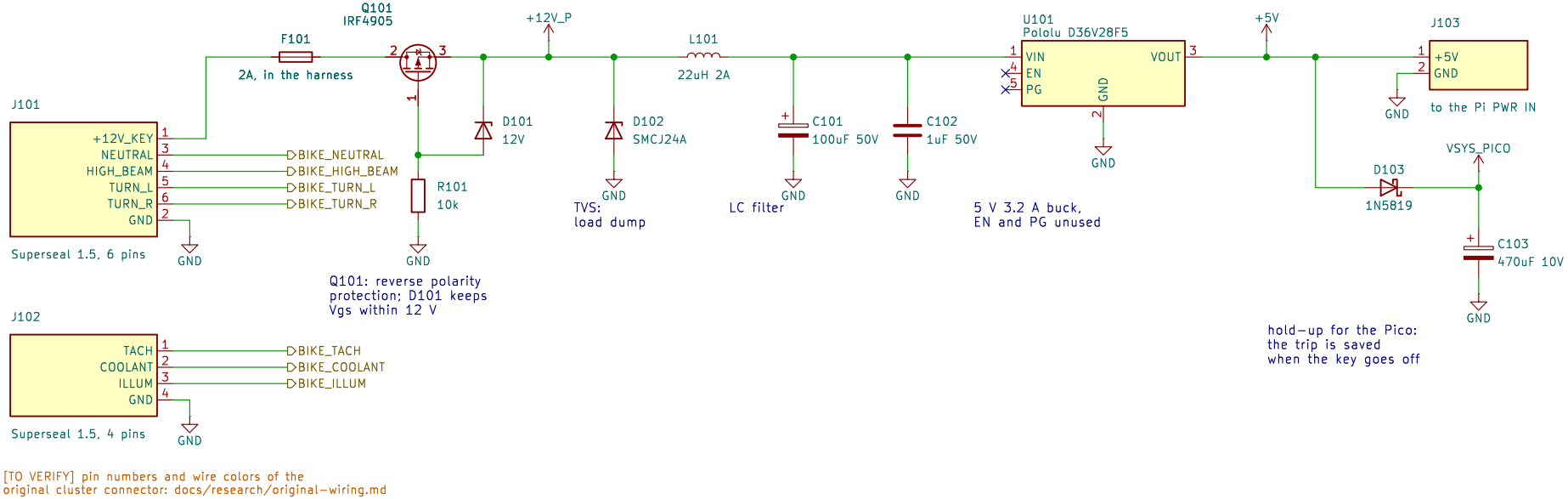
tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

Overview: block diagram

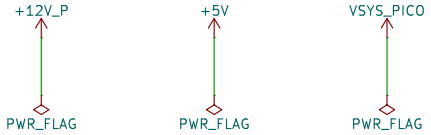
SHEET	REV	DATE	SIZE
0 / 8	0.2	2026-10-21	A4

Bike connectors, protection and 5 V supply

+12V_KEY is the switched 12 V of the bike. Chain: 2 A fuse in the harness, P-MOSFET against reversed polarity, 24 V TVS for the load dump, LC filter, 5 V buck converter for the Pi, the backlight and the Pico. The Pico runs from VSYS_PICO through D103: C103 keeps it alive long enough to save the trip.



[TO VERIFY] pin numbers and wire colors of the original cluster connector: docs/research/original-wiring.md



Power flags: the ERC learns that these rails are supplied on this sheet.

github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand

tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

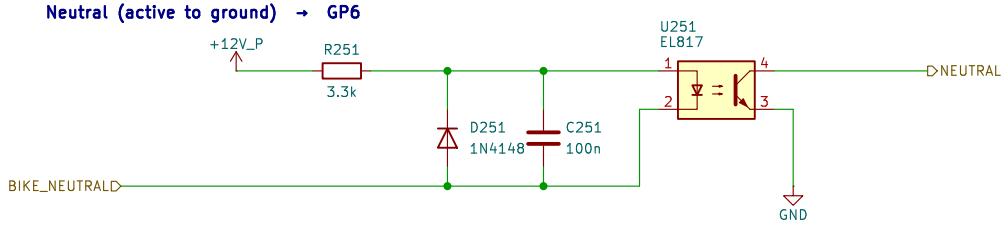
Power supply and protection

SHEET	REV	DATE	SIZE
1 / 8	0.2	2026-10-21	A4

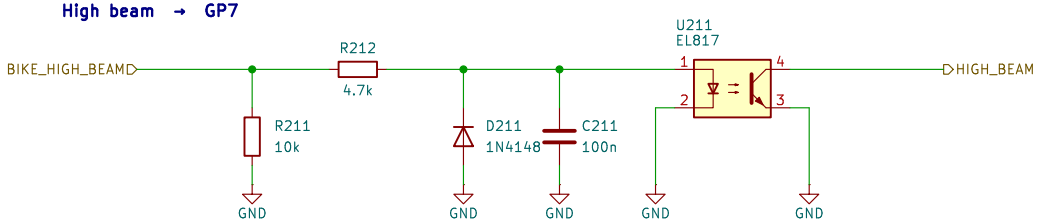
Warning light inputs

High beam, turn signals and illumination are active at +12 V: 4.7 kΩ and the EL817 LED to ground, 10 kΩ to ground so an open wire reads off. Neutral is active to ground: the gearbox switch grounds the wire, so the LED is fed from +12V_P. Outputs are active low, with the Pico's internal pull-ups.

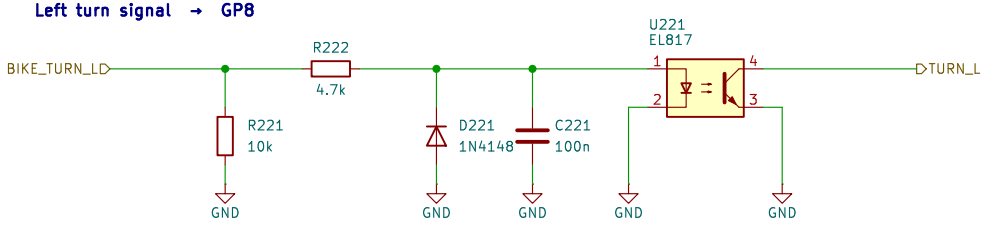
Neutral (active to ground) → GP6



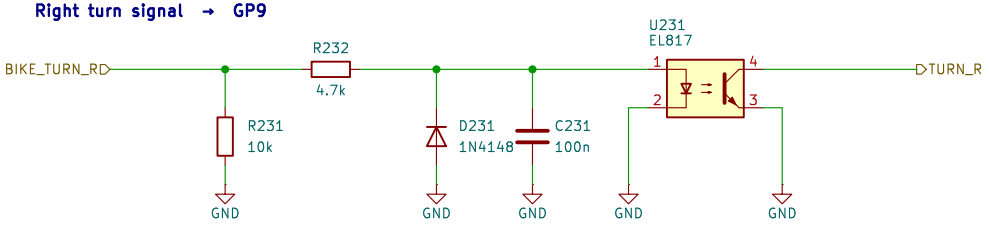
High beam → GP7



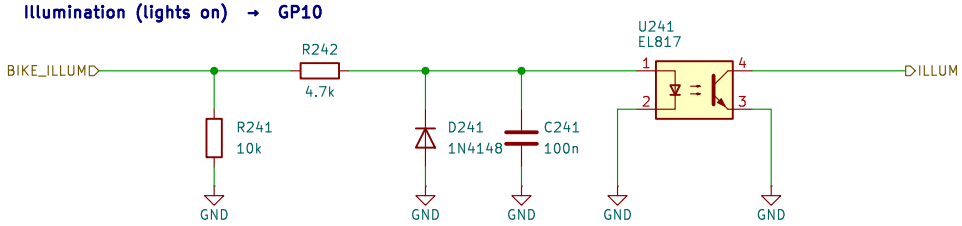
Left turn signal → GP8



Right turn signal → GP9



Illumination (lights on) → GP10



Channel by channel:
R2x2 4.7 kΩ: about 2 mA in the LED at 12 V
D2x1: protects the LED from reverse voltage
C2x1: filters spikes on the bike wire
R2x1 10 kΩ: an open wire reads off

The EL817 collector pulls the GPIO low when the warning light is on.

[TO VERIFY] neutral logic and wire colors, with the bike measurements: docs/research/original-wiring.md

github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand

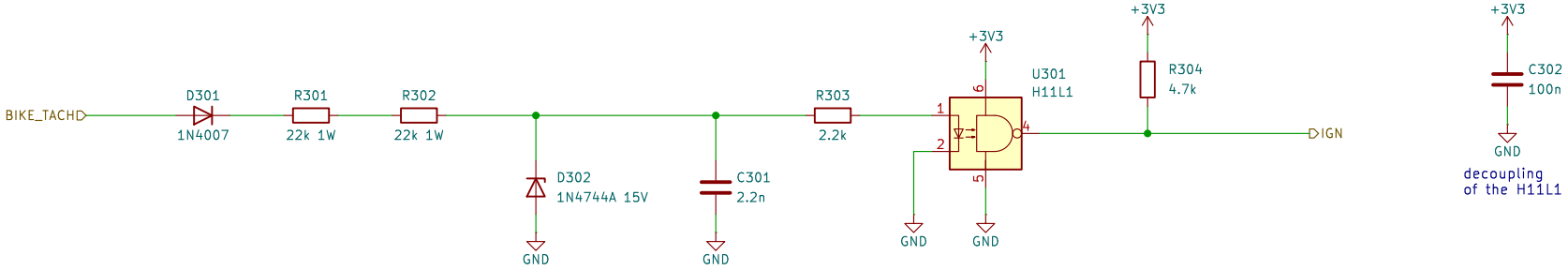
tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

Warning light inputs (optocouplers)

SHEET	REV	DATE	SIZE
2 / 8	0.2	2026-10-21	A4

Tachometer input

The coil negative swings by a few hundred volts at every spark.
D301 passes the positive spikes, R301 and R302 limit the current, D302 and C301 clamp and filter them; the H11L1 Schmitt trigger gives the Pico a clean edge.



passes only the positive spikes

2 × 22 kΩ 1 W: limit the current

clamp at 15 V, filter

Schmitt trigger: clean edges

IGN → GP3, active low

[TO VERIFY] tachometer wire and its waveform, with the oscilloscope: docs/research/original-wiring.md

github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand

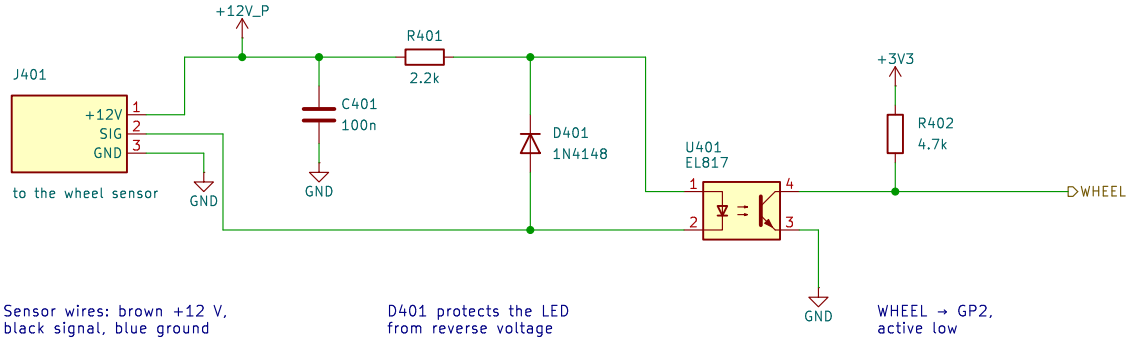
tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

Tachometer (coil negative)

SHEET	REV	DATE	SIZE
3 / 8	0.2	2026-10-21	A4

Wheel speed sensor

LJ12A3-4-Z/BX inductive sensor, NPN normally open: in front of a bolt its output goes to ground.
The sensor sinks the current of the EL817 LED, which is fed from +12V_P through R401;
R402 pulls the output up for fast edges.



Sensor wires: brown +12 V, black signal, blue ground

D401 protects the LED from reverse voltage

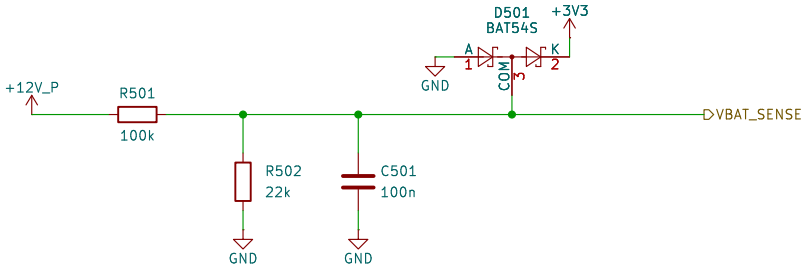
WHEEL → GP2, active low

github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand			
tdm-dash sensor module · Yamaha TDM 850 (3VD)			CERN-OHL-P-2.0
Wheel sensor (LJ12A3)			
SHEET 4 / 8	REV 0.2	DATE 2026-10-21	SIZE A4

Analog inputs

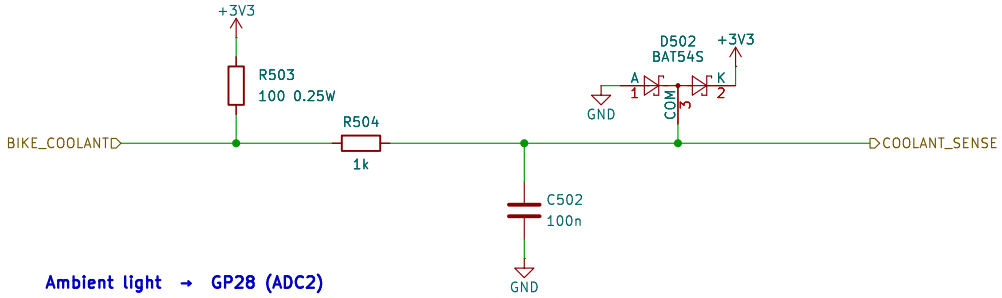
Each input reaches its Pico ADC pin through a filter capacitor and a BAT54S that clamps it between ground and 3.3 V, so a spike on the bike wiring cannot reach the GPIO.

Battery voltage → GP26 (ADC0)



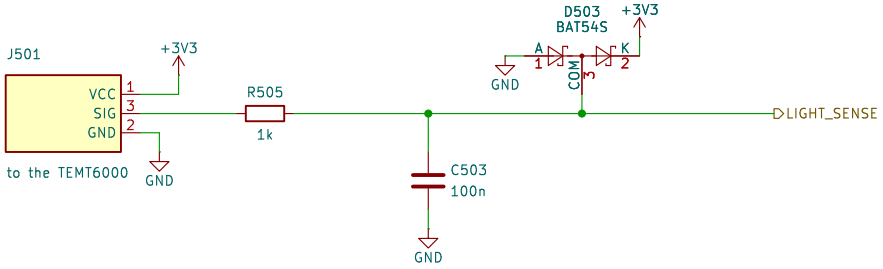
100k / 22k: 18.3 V at the battery gives 3.3 V, the top of the ADC scale

Coolant temperature → GP27 (ADC1)



The sender is an NTC to ground in the engine: R503 pulls it up to 3.3 V, R504 limits the current into the clamp

Ambient light → GP28 (ADC2)



TEMT6000 module behind the window, supplied at 3.3 V

github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand

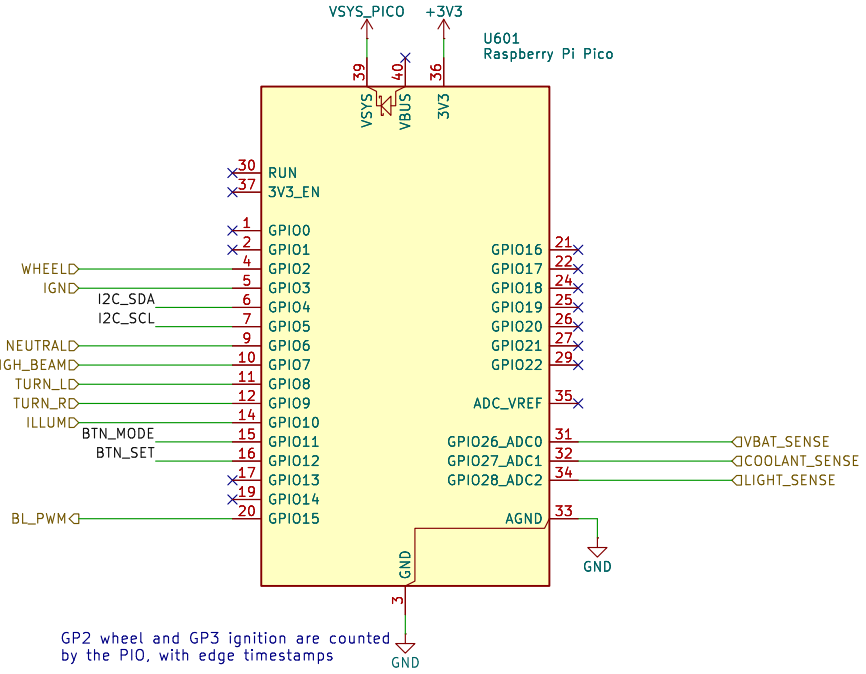
tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

Analog inputs

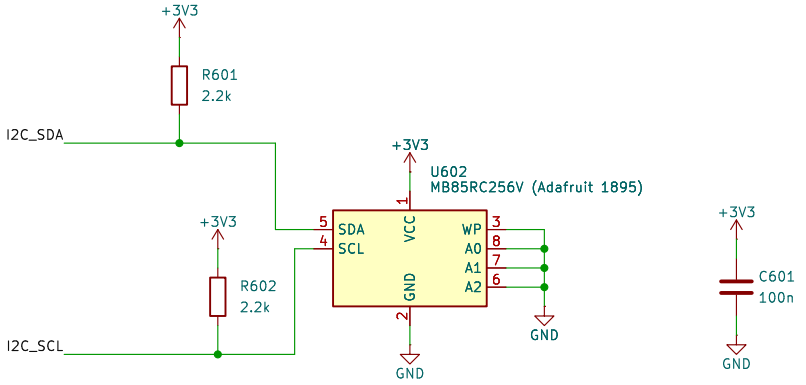
SHEET	REV	DATE	SIZE
5 / 8	0.2	2026-10-21	A4

Raspberry Pi Pico, FRAM and buttons

Raspberry Pi Pico or Pico 2: same pinout, same firmware. Unused pins are left open.
The FRAM keeps the odometer (docs/adr/0004); MODE and SET are filtered against bounce and discharges.

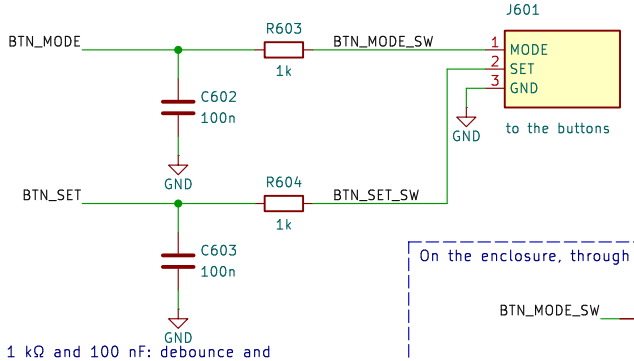


GP2 wheel and GP3 ignition are counted by the PIO, with edge timestamps

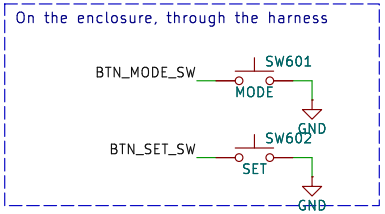


I²C at 1 MHz, address 0x50 (A0–A2 to ground)

[TO VERIFY] header pin order of the breakout



1 kΩ and 100 nF: debounce and protection against discharges



github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand

tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

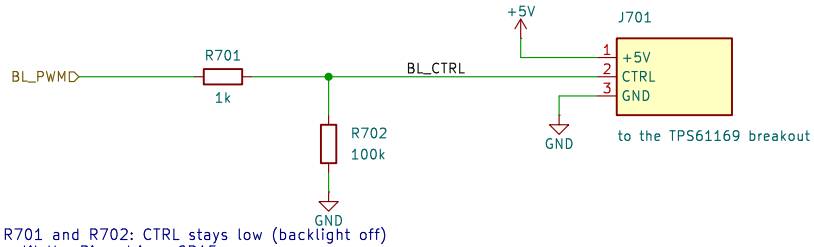
Pico, FRAM and buttons

SHEET	REV	DATE	SIZE
6 / 8	0.2	2026-10-21	A4

Backlight and USB link to the Pi

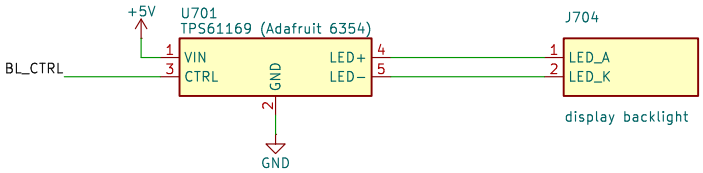
The Pico dims the backlight with 25 kHz PWM on GP15. R702 holds CTRL low until the firmware runs, so the display stays dark if the Pi or the Pico hang (fail-dark, docs/adr/0006).

Display backlight



R701 and R702: CTRL stays low (backlight off) until the Pico drives GP15

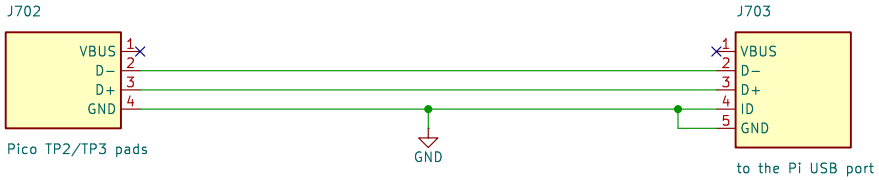
Off the board: TPS61169 breakout next to the Kippah, wired to J701 (backlight harness, chapter 05)



Newhaven panel: LEDs at 180 mA, 9.3 V

[TO VERIFY] breakout pins, and the panel LED pins on the Kippah: panel datasheet, chapter 05

USB between the Pico and the Pi Zero 2 W



Short twisted pair with a clip-on ferrite.
VBUS is not connected: the Pico is supplied from VSYS (sheet 1).
ID is tied to GND in the plug, so the Pi works as USB host.

The display itself connects to the Pi through the Adafruit DPI Kippah (40-pin FFC), not through this board.

github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand

tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

Backlight and link to the Pi

SHEET	REV	DATE	SIZE
7 / 8	0.2	2026-10-21	A4

Backup LEDs

Wired straight to the bike's 12 V lines, so they light up even with the electronics off.
The turn signals use the pair of wires of the original indicator: the flashing side feeds its LED,
the other side goes to ground through its bulbs. The 1N4148 protects each LED from the reverse 12 V.

Left turn signal

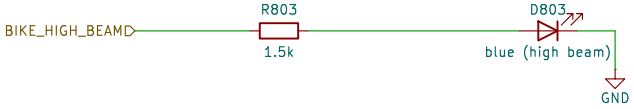


Current flows from the flashing side,
through its LED, to the other side,
which the bulbs hold near ground.

Right turn signal



High beam



Neutral



The gearbox switch grounds the neutral wire.

github.com/frascella-dev/tdm-dash · generated by hardware/kicad/generate.py, do not edit by hand

tdm-dash sensor module · Yamaha TDM 850 (3VD) CERN-OHL-P-2.0

Backup LEDs

SHEET	REV	DATE	SIZE
8 / 8	0.2	2026-10-21	A4